

4	(a)	When a charged particle is travelling in a magnetic field, it experiences a <u>magnetic force that is always perpendicular to its velocity</u> (and the magnetic field lines). Since the charged particle is travelling perpendicular to the uniform magnetic field, and that the <u>resultant force only consists of the magnetic force</u>, the particle travels in a circular path.	B1 B1
	(b)	centripetal force provided by magnetic force	M1 M1

		$mr\omega^2 = Bqv$ $m\omega^2 = Bq\left(\frac{v}{r}\right)$ $m\omega^2 = Bq\omega$ $\omega = \frac{Bq}{m}$ $\frac{2\pi}{T} = \frac{Bq}{m}$ $T = \frac{2\pi m}{Bq}$	A1
	(b)	magnetic force provides for centripetal force $Bqv = \frac{mv^2}{r}$ $q = \frac{mv}{Br} = \frac{(4.5 \times 10^{-26})(4.8 \times 10^5)}{(0.15)\left(\frac{0.60}{2}\right)}$ $= 4.8 \times 10^{-19} \text{ C}$	M1 A1

5	(a)	(i)	Hypothetical gas obeys equation of state $pV = nRT$ (perfectly at all	P1
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